SNEHAL SINHA

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EDUCATION

GEORGIA INSTITUTE OF TECHNOLOGY, Scheller College of Business — **Atlanta, GA**

Master of Science in Quantitative and Computational Finance — *Jan 2026 – May 2027*

- **GPA: 4.00/4.00**; Coursework: Stochastic Processes for Asset Pricing, Computational Data Analysis, Computational Statistics and Optimization, Numerical Methods in Finance, Fixed Income Securities, Derivative Securities, AI in Finance
- **Position:** Awarded Graduate Assistantship (GA) for AI in Finance course

Indian Institute of Technology (IIT) Delhi — **New Delhi, India**

Bachelor of Technology, Civil Engineering, GPA: 8.37/10.00 (Top 7%) — *May 2018 – May 2022*

EXPERIENCE

BlackRock — **Gurugram, India**

Associate, Portfolio Risk, Aladdin Financial Engineering — *Jan 2025 – Dec 2025*

- Implemented and productionized a Merton structural-credit model that decomposes private credit risk into explainable risk factors, leveraging Monte Carlo simulation for estimating the probability of default. Designed an alternative volatility-calibration algorithm that cut simulation runtimes 60%+ and reduced production run-time incidents
- Developed and validated a stochastic cashflow-forecasting model used to simulate capital flows for commitment-based investment vehicles; performed back-testing, designed test cases, stress tested and benchmarked against industry models
- Enabled 40% efficiency by incorporating variance reduction techniques for simulations in portfolio risk models
- Served as SME for factor-based models covering $300B+ AUM across multiple asset classes, resolving client escalations on VaR, stress testing, and risk-decomposition issues; recognized in an internal newsletter for quick turnaround
- Supported deployment of BlackRock Capital Market Assumptions (CMAs) used by institutional investors worldwide to track macro-economic trends by providing distribution of risk breakdown of private investments across geographies and sectors

Analyst, Portfolio Risk, Aladdin Financial Engineering — *Aug 2022 – Dec 2024*

- Led validation efforts for portfolio weight data-store application as part of major infrastructure migration to micro-services, performing regression testing for millions of investment positions spanning diverse asset class holdings
- Built an automated Python-based framework for systematic integration, benchmarking and back-testing of Equity risk models
- Achieved 90% reduction in run-times for a large-scale position reconciliation API-based tool; later adopted by multiple teams

PROJECTS & RESEARCH EXPERIENCE

Systematic Intraday Long-Volatility Trading Strategy — **Atlanta, GA**

- Built an intraday delta-neutral long-volatility strategy on Nifty 50 options trading the realized–implied variance spread; back tested to +10.1% over a two-month expiry cycle (Sharpe 1.86, profit factor 2.27, 6.1% max drawdown, 63%-win rate)
- Leveraged volatility surface signals (25Δ butterfly, put-call skew, skew slope) and the combined gamma-theta P&L identity to time entries, then conditioned strike selection, inverse-IV sizing and a staged exit ladder based on the IV regime
- Applied small-sample inference (Lo-adjusted Sharpe, 10,000-resample bootstrap) to flag the result as hypothesis-generating

Factor decomposition of momentum crowding — **Atlanta, GA**

- Extended "Comomentum" crowding signal from high-frequency abnormal return correlations across momentum-sorted deciles, using Fama-MacBeth and panel regressions to forecast time-varying momentum returns and predict strategy crashes
- Decomposed the crowding measure against alternative momentum factors (residual/idiosyncratic, industry, 52-week-high) via cross-sectional regressions to isolate overreaction- vs. underreaction-driven return components

Equity Sector Rotation Trading strategy using ML — **Atlanta,GA**

- Built an FOMC "hawkishness" signal via LLM inference on FOMC meeting minutes; validated it against Fed Funds moves (83% directional hit rate) and ran Fama-French sector rate-sensitivity regressions to drive a long-short sector-rotation strategy

ACHIEVEMENTS & INTERESTS

IMC Prosperity 4 Trading competition: Top 5% in manual trading round by constructing a synthetic put via put-call-parity/static replication, arbitraging the pricing spread

World Cube Association (WCA): Official 3×3 14.02s solve (national top 10%); debutant podium (ID: 2024SINH05)

TECHNICAL SKILLS & CERTIFICATIONS

Certifications: Qualified CFA Level 1 and FRM Level 1 examinations

Programming: Python (NumPy, Pandas, SciPy, PyTorch), C++, R, Java, SQL, Excel VBA